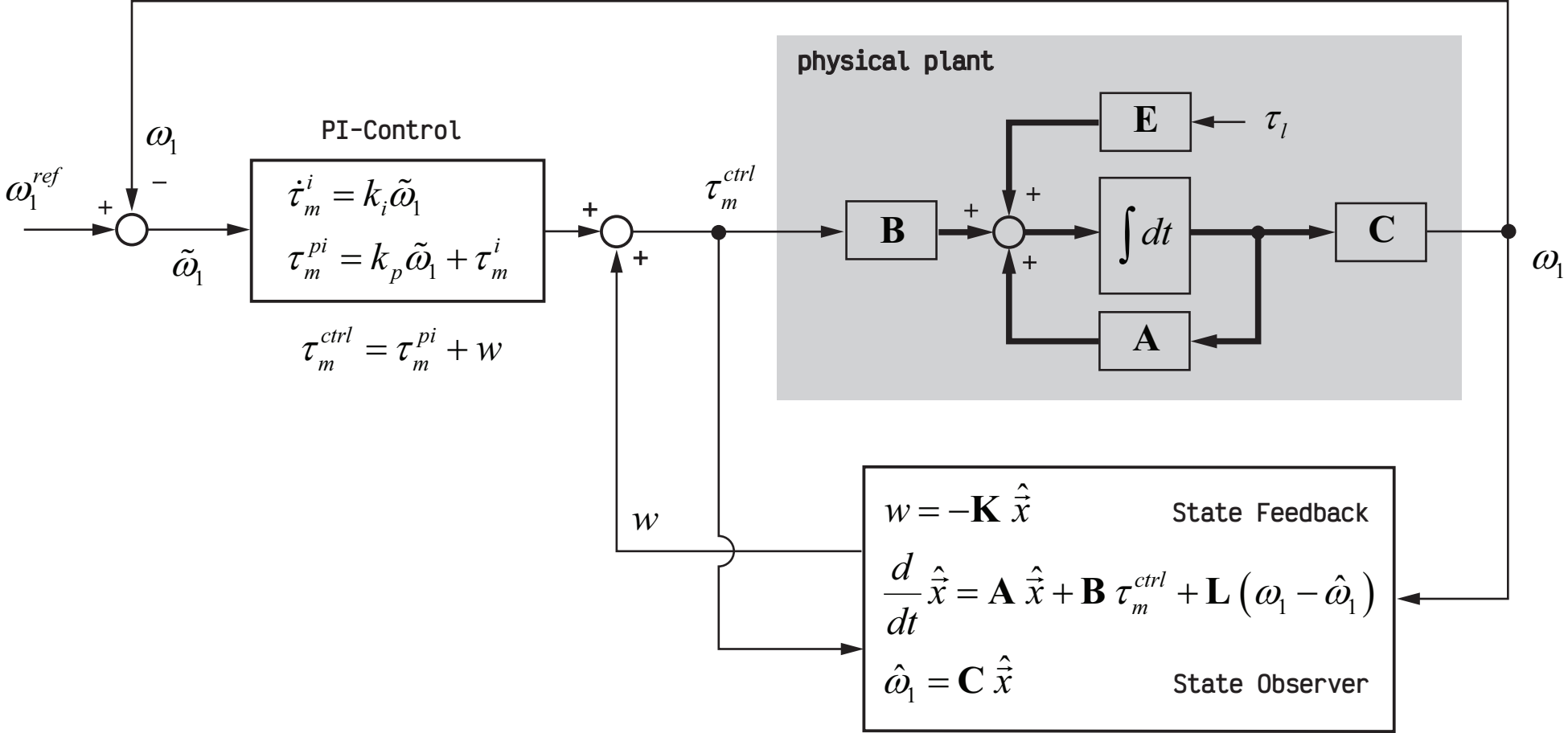
physical plant
PI-Control
ω_1
ω_1^{ref}
$\tilde{\omega}_1$
$\dot{\tau}_m^i = k_i \tilde{\omega}_1$
$\tau_m^{pi} = k_p \tilde{\omega}_1 + \tau_m^i$
$\tau_m^{ctrl} = \tau_m^{pi} + w$
τ_m^{ctrl}
E
τ_l
B
$\int dt$
C
A
ω_1
w
$w = -\mathbf{K}\,\hat{\vec{x}}$
State Feedback
$\frac{d}{dt}\hat{\vec{x}} = \mathbf{A}\,\hat{\vec{x}} + \mathbf{B}\,\tau_m^{ctrl} + \mathbf{L}\left(\omega_1 - \hat{\omega}_1\right)$
$\hat{\omega}_1 = \mathbf{C}\,\hat{\vec{x}}$
State Observer